MGMT 59000 Final Project

Computing for Analytics

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Introduction:

Our dataset is from Uber, which is a large taxi company. Dataset is at a trip level, giving information about the pickup date-time, pickup location, drop-off location fare amount, and the number of passengers on the trip. The dataset contains data from the start of 2009 till mid-2015. For the detailed data dictionary, see below:

- **key:** a unique identifier of each trip (It's the same as the pickup date and time)
- **fare_amount:** The cost of each trip in USD
- **pickup_datetime:** Date and time when the meter was engaged
- **passenger_count:** the number of passengers in the vehicle (driver entered value)
- **pickup_longitude:** the longitude where the meter was engaged
- **pickup_latitude:** the latitude where the meter was engaged
- **dropoff_latitude:** the longitude where the meter was disengaged
- **dropoff_latitude:** the latitude where the meter was disengaged

For our analysis, we have used data for 2015 to consider the most recent data available. As part of our analysis, we have used several graph theory, optimization, and machine learning algorithms to gather insights into the data as mentioned above. Following techniques have been used to as part of the project from our course to gather insights:

- **Random Forest Regression** to predict fare for future trips
- **K-Means Clustering** Algorithm to identify trip clusters
- **Weakly Connected Components** for directed graphs
- **Minimum Vertex Cover** Optimization
- **Bellman Ford Algorithm** for Shortest Path

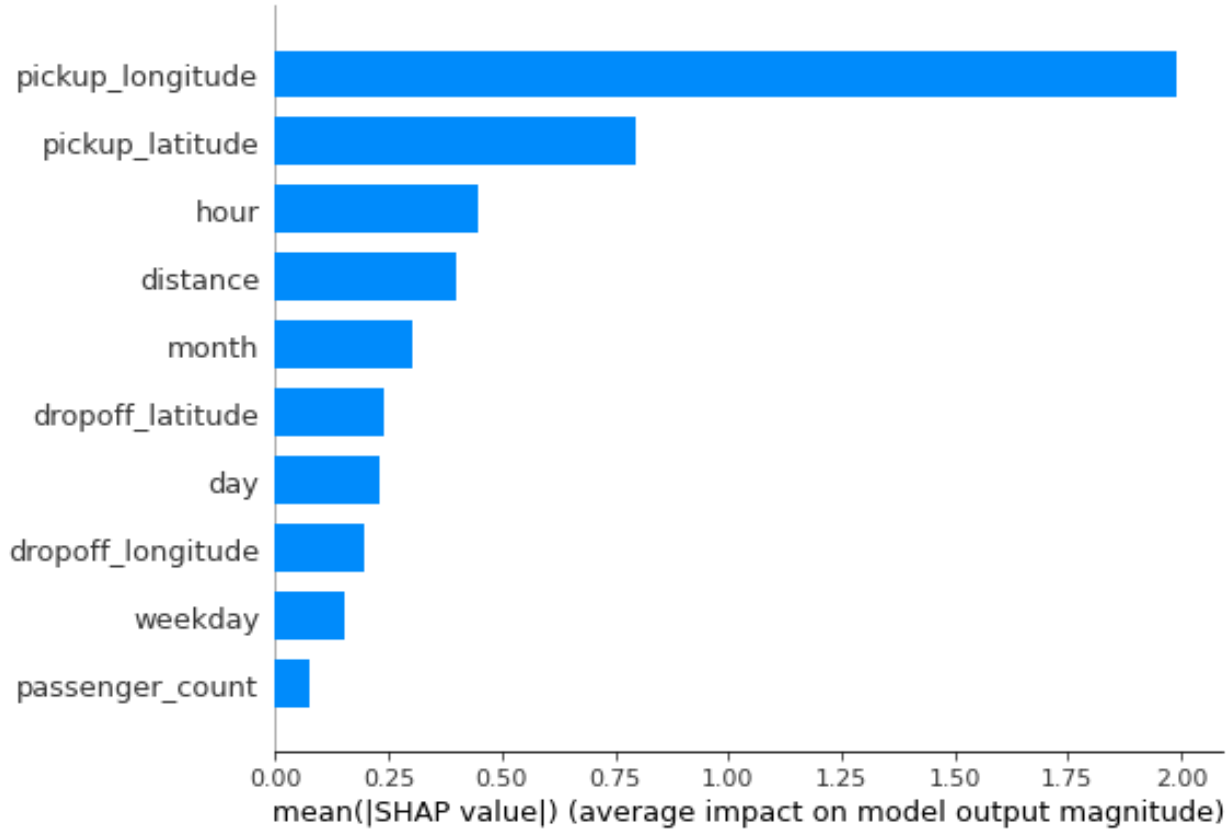
Random Forest Regression:

We have used the uber data to create a machine learning model to predict fare for future rides. For our analysis, we used Random Forest as our algorithm as it has been used widely for such kinds of analysis and our analysis also gave out some great results. We did some feature engineering at our end to allow algorithm to learn in the best way possible. We have created below additional columns:

- **Hour:** Hour during which the ride meter was engaged
- **Day:** Day of the week when the meter was engaged
- **Month:** Month in which the meter was engaged
- **Weekday:** Which day was it i.e., Sunday, Monday etc.
- **Distance:** Distance between pickup and drop off location

We also used Grid Search Cross Validation to get the best parameter values for our model with the lowest Root Mean Square Error. We have optimized parameters such as Min Samples Split, Bootstrap, Min Samples Leaf, etc. We were able to achieve an RMSE Score of 9.92 post grid search optimization which is well within expectation for the quality of data we have.

From our analysis, we got the importance of different variables for fare prediction. Please see below plot for the same:

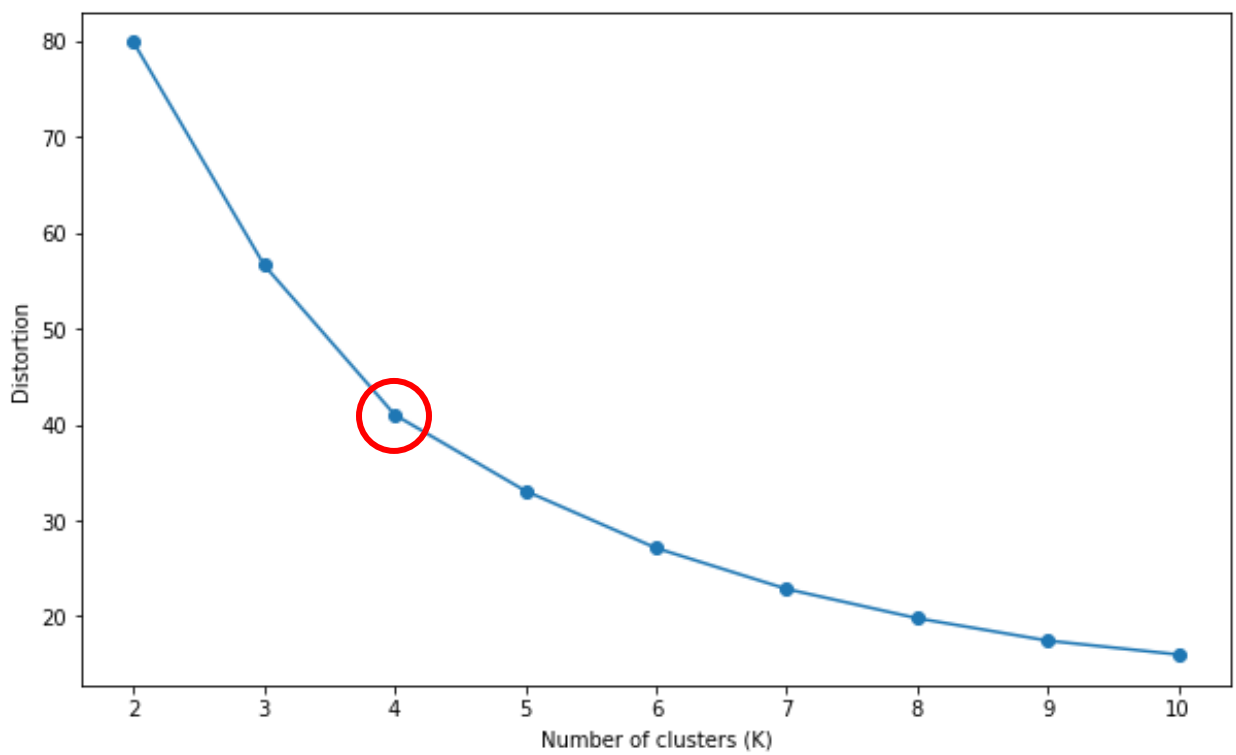


Based on the above plot, we see that Pickup location is the most crucial variable in predicting the fare for a ride. This could be because, for example, if someone usually takes an uber from a densely populated place and a lot of people tend to book cabs there, then the fare is expected to be high due to demand. The next feature which is essential according to the above plot is the hour – Hour is an important variable in predicting fare as there are several peak times when most people book cabs, like in the morning to reach the office and when going home from the office in the evening. Such times can expect a higher fare, and rest times, we can expect comparatively lower fares. Distance is another crucial variable, as the fare is directly proportional to your distance. The month is an essential variable, as you can expect the fare to go up considerably during festival or vacation months.

K-Means Clustering:

We have used the K-means clustering algorithm to identify the best cluster for trips that drivers can use to determine which pick-up and drop-off combinations are best to achieve a particular fare. We have used two variables as part of this analysis, i.e., passenger_count and distance. We had to feature engineering additionally for this analysis which is scaling in our case. K-Means requires all the variables in our data to be scaled (Min Max Scaler in our case) at the same level as the algorithm calculates Euclidean distance. Hence, scaling ensures certain variables don't have most of the effect when creating clusters.

We created an elbow curve to identify the best K for our analysis. Please find below the plot for the same:



As we can see above, the elbow is coming at 4, i.e., it would make sense to keep four as K for our analysis to get maximum variability.

Next, we fit our K-Means algorithm using four as K and identify the clusters on the entire data. Below is a quick summary of the Cluster with mean distance and fare amount along with Min and Max.

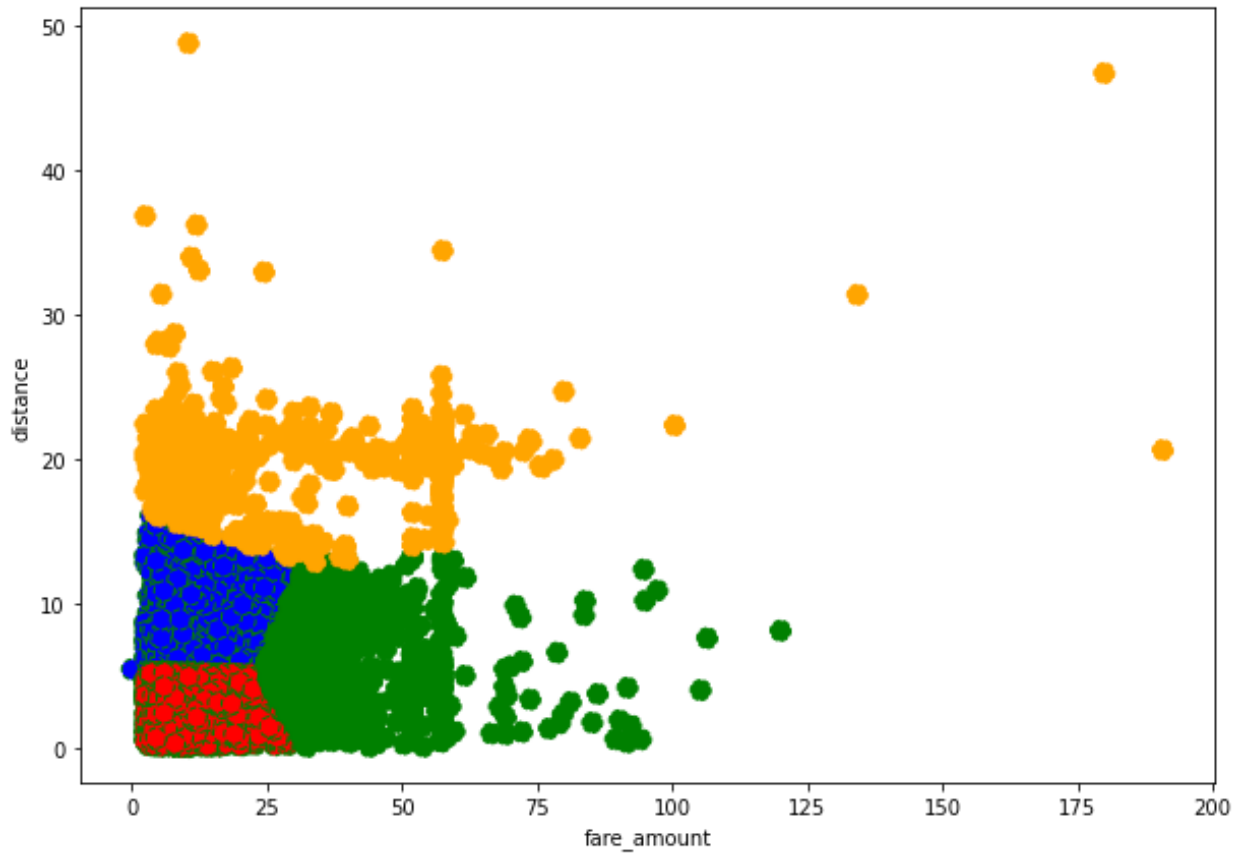
	Fare Amount			Distance			
Cluster	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**	**Fare / Distance**
0	10.52	0.01	31.00	7.77	5.15	16.04	1.35
1	10.06	2.50	31.00	2.61	0.01	5.22	3.86
2	41.00	25.00	120.08	5.77	0.04	13.30	7.10
3	32.98	2.50	190.83	19.98	12.83	48.76	1.65

Based on the above table, we can observe that 3rd Cluster has a high mean distance and fare amount, but overall fare per distance is not that high. This could be due to trips around the suburban areas where fare amounts can't be high.

2nd Cluster is the one with the highest mean distance and fare amount and provides high fare/distance. These trips could be pretty lucrative for drivers and can help them earn higher compared to other drivers

0 and 1st Cluster have similar mean fare amounts but different mean distances. These could be due to several factors like the pickup location, time of pickup, etc.; hence, more research is needed on why we see such behavior.

We also created a Scatter Plot between Fare Amount and Distance to show the clusters graphically. Please find the plot below:



From the scatter plot, we can see an evident variation among clusters. Overall, K-Means can help drivers optimize their trips and help Uber generate more revenue.

Weakly Connected Components:

We have taken one month of 2015 data(Jan 2015) for the network to save computational time. A weighted directed graph is created using source nodes as "Pickup_Locations" and target nodes as "Dropoff_Locations", with weights of edges as both 'distance'(average) and 'fare amount' (average) between the locations. Since it is a directed graph, we have found out the "Weakly connected components" I.e. all the vertices are connected ignoring the direction of edges. This helps us identify the undirected routes from one location to another(since a trip can be made in an opposite direction as well).

Directed Graph Properties:

Number of Nodes: 3257

Number of Edges: 2069

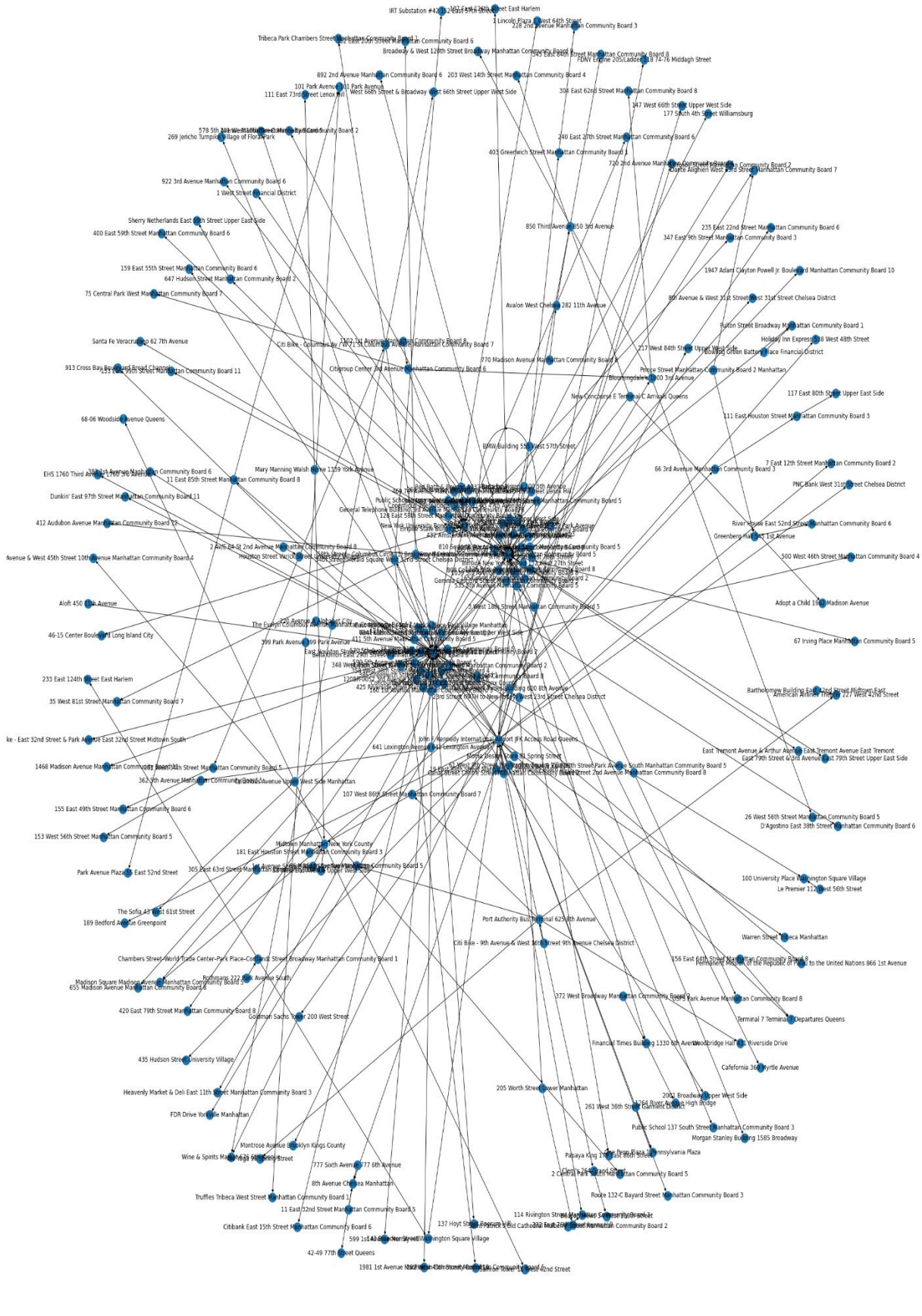
Number of Weakly Connected Components : 1191

Number of Nodes in the largest weakly connected Component:224

Number of Edges in the largest weakly connected Component:226

We have 224 pick_up and drop_off locations in the largest connected components below, implying that the trips have been largely concentrated within these locations for the given time period.

Following is the network for the largest weakly connected component(G0):


IRT Substation #42 East 57th Street
1 Lincoln Plaza West 64th Street
228 2nd Avenue Manhattan Community Board 3
Tribeca Park Chambers Street Manhattan Community Board 1
Broadway & West 120th Street Broadway Manhattan Community Board 9
245 East 84th Street Manhattan Community Board 8
FDNY Engine 205/Ladder 118 74-76 Middagh Street
892 2nd Avenue Manhattan Community Board 6
203 West 14th Street Manhattan Community Board 4
101 Park Avenue 41 Park Avenue
111 East 73rd Street Lenox Hill
West 66th Street & Broadway West 66th Street Upper West Side
304 East 62nd Street Manhattan Community Board 8
147 West 66th Street Upper West Side
177 South 4th Street Williamsburg
578 5th Avenue Manhattan Community Board 5
269 Jericho Turnpike Village of Floral Park
240 East 27th Street Manhattan Community Board 6
403 Greenwich Street Manhattan Community Board 1
730 2nd Avenue Manhattan Community Board 2
Carte Alighen West 23rd Street Manhattan Community Board 7
922 3rd Avenue Manhattan Community Board 6
1 Wall Street Financial District
Sherry Netherlands East 59th Street Upper East Side
400 East 59th Street Manhattan Community Board 6
850 Third Avenue 850 3rd Avenue
235 East 22nd Street Manhattan Community Board 6
347 East 9th Street Manhattan Community Board 3
159 East 55th Street Manhattan Community Board 6
647 Hudson Street Manhattan Community Board 2
1947 Adam Clayton Powell Jr. Boulevard Manhattan Community Board 10
75 Central Park West Manhattan Community Board 7
Avalon West Chelsea 282 11th Avenue
801 Avenue & West 31st Street West 31st Street Chelsea District
Fulton Street Broadway Manhattan Community Board 1
Holiday Inn Express 138 West 48th Street
Santa Fe Veracruzano 62 7th Avenue
Citi Bike - Columbus Ave 71 St. Columbus Avenue Manhattan Community Board 7
217 West 84th Street Upper West Side
Belgian Green Battery Place Financial District
770 Madison Avenue Manhattan Community Board 8
913 Cross Bay Boulevard Broad Channel
Citigroup Center 2nd Avenue Manhattan Community Board 6
Prince Street Manhattan Community Board 2 Manhattan
Bloomingdale's 1000 3rd Avenue
New Concourse E Terminal C Arrivals Queens
117 East 80th Street Upper East Side
111 East Houston Street Manhattan Community Board 3
68-06 Woodside Avenue Queens
BMW Building 555 West 57th Street
7 East 12th Street Manhattan Community Board 2
66 3rd Avenue Manhattan Community Board 3
Mary Manning Walsh Nurse 1359 York Avenue
EHS 1760 Third Avenue 1760 3rd Avenue
11 East 85th Street Manhattan Community Board 8
PNC Bank West 31st Street Chelsea District
Dunkin' East 97th Street Manhattan Community Board 11
412 Audubon Avenue Manhattan Community Board 12
River House East 52nd Street Manhattan Community Board 6
Greenberg Hall 145 1st Avenue
2 Ave 84 St 2nd Avenue Manhattan Community Board 8
Avenue & West 45th Street 10th Avenue Manhattan Community Board 4
500 West 44th Street Manhattan Community Board 4
Aloft 450 10th Avenue
Adopt a Child 1950 Madison Avenue
46-15 Center Boulevard Long Island City
67 Irving Place Manhattan Community Board 5
233 East 124th Street East Harlem
35 West 81st Street Manhattan Community Board 7
Bartholomew Building East 42nd Street Midtown East
American Airlines Tower 227 West 42nd Street
John F. Kennedy International Airport JFK Access Road Queens
East Tremont Avenue & Arthur Avenue East Tremont Avenue East Tremont
East 79th Street & 3rd Avenue East 79th Street Upper East Side
1468 Madison Avenue Manhattan Community Board 5
362 9th Avenue Manhattan Community Board 4
107 West 86th Street Manhattan Community Board 7
155 East 49th Street Manhattan Community Board 6
26 West 56th Street Manhattan Community Board 5
D'Agostino East 38th Street Manhattan Community Board 6
153 West 56th Street Manhattan Community Board 5
Midtown Manhattan New York County
181 East Houston Street Manhattan Community Board 3
Park Avenue Plaza 55 East 52nd Street
100 University Place Washington Square Village
Le Premier 112 West 56th Street
The Sofia 43 West 61st Street
189 Bedford Avenue Greenpoint
Port Authority Bus Terminal 625 8th Avenue
Warren Street Tribeca Manhattan
Citi Bike - 9th Avenue & West 36th Street 9th Avenue Chelsea District
Chambers Street-World Trade Center-Park Place-Cortlandt Street Broadway Manhattan Community Board 1
156 East 64th Street Manhattan Community Board 8
Permanent Mission of the Republic of Peru to the United Nations 866 1st Avenue
Madison Square Madison Avenue Manhattan Community Board 5
655 Madison Avenue Manhattan Community Board 6
Rothmans 222 Park Avenue South
372 West Broadway Manhattan Community Board 2
Park Avenue Manhattan Community Board 8
420 East 79th Street Manhattan Community Board 8
Goldman Sachs Tower 200 West Street
Terminal 7 Terminal 7 Departures Queens
Financial Times Building 1330 6th Avenue
Woodbridge Hall 601 Riverside Drive
435 Hudson Street University Village
Cafefornia 364 Myrtle Avenue
205 Worth Street Tower Manhattan
Heavenly Market & Deli East 11th Street Manhattan Community Board 3
2002 Broadway Upper West Side
261 West 36th Street Garment District
FDR Drive Yorkville Manhattan
Public School 137 South Street Manhattan Community Board 3
Morgan Stanley Building 1585 Broadway
Montrose Avenue Brooklyn Kings County
Papaya King 1123 East 86th Street
Penn Plaza Pennsylvania Plaza
Wine & Spirits Market 55 Spring Street
777 Sixth Avenue 777 6th Avenue
2 Central Park South Manhattan Community Board 5
8th Avenue Chelsea Manhattan
Truffles Tribeca West Street Manhattan Community Board 1
Route 132-C Bayard Street Manhattan Community Board 3
11 East 32nd Street Manhattan Community Board 5
114 Rivington Street
Citibank East 15th Street Manhattan Community Board 6
137 Hoyt Street
Bleecker Street Manhattan Community Board 2
599 1st Avenue Washington Square Village
42-49 77th Street Queens
1981 1st Avenue
West 42nd Street

Minimum Vertex Cover Optimization

We performed a minimum vertex cover optimization on the largest connected component subgraph(G0). Out of 224 nodes, we obtained 52 locations that give us the minimum vertex cover. These 52 locations cover all the paths between the 224 locations(having at least one location as a part of each route)

The Uber company can focus on these 52 locations as pick up or drop off for maximizing the number of trips and hence fare amount i.e. since these locations cover all the routes within the given connected component. The following locations constitute the minimum vertex cover:

	Vertex Cover Locations
0	100 University Place Washington Square Village
1	111 East 73rd Street Lenox Hill
2	111 East Houston Street Manhattan Community Bo...
3	11 East 85th Street Manhattan Community Board 8
4	153 West 56th Street Manhattan Community Board 5
5	155 East 49th Street Manhattan Community Board 6
6	156 East 64th Street Manhattan Community Board 8
7	235 East 22nd Street Manhattan Community Board 6
8	23rd Street (PATH to New Jersey) West 23rd Str...
9	261 West 36th Street Garment District
10	305 East 63rd Street Manhattan Community Board 8
11	332 East 76th Street Lenox Hill
12	345 East 84th Street Manhattan Community Board 8
13	348 West 49th Street Manhattan Community Board 4
14	34th Street–Herald Square West 32nd Street Che...
15	399 Park Avenue 399 Park Avenue
16	403 Greenwich Street Manhattan Community Board 1
17	432 Park Avenue 432 Park Avenue
18	535 5th Avenue Manhattan Community Board 5
19	59th Street Columbus Circle (1) Broadway Man...
20	647 Hudson Street Manhattan Community Board 2
21	86 Madison Avenue Manhattan Community Board 5
22	86th Street 2nd Avenue Manhattan Community Boa...
23	8th Avenue Chelsea Manhattan
24	90 Park Avenue 90 Park Avenue

25	Avalon West Chelsea 282 11th Avenue
26	Bartholomew Building East 42nd Street Midtown ...
27	Beacon Mews 34 West 139th Street
28	Bella Union East 29th Street Manhattan Communi...
29	Bloomingdale's 1000 3rd Avenue
30	Chambers Street–World Trade Center–Park Place–...
31	Citigroup Center 3rd Avenue Manhattan Communit...
32	FDR Drive Yorkville Manhattan
33	Financial Times Building 1330 6th Avenue
34	Fulton Street Broadway Manhattan Community Boa...
35	General Telephone Building 3rd Avenue Manhatta...
36	Grand Central Terminal Park Avenue Manhattan C...
37	Houston Street Varick Street University Village
38	John F. Kennedy International Airport JFK Acce...
39	Midtown Manhattan New York County
40	New Concourse E Terminal C Arrivals Queens
41	New York University Bond Street Manhattan Comm...
42	Pennsylvania Station 8th Avenue Chelsea District
43	Port Authority Bus Terminal 625 8th Avenue
44	Prince Street Manhattan Community Board 2 Manh...
45	River House East 52nd Street Manhattan Communi...
46	Saint Patrick's Old Cathedral Mulberry Street ...
47	The Evelyn Columbus Avenue Manhattan Community...
48	The New York Times Building 620 8th Avenue
49	The Sofia 43 West 61st Street
50	Tribeca Park Chambers Street Manhattan Communi...
51	Wine & Spirits Market 676 6th Avenue

Bellman Ford Algorithm for Shortest Path

Suppose an Uber driver is waiting at “Grand Central Terminal” for passenger pickups. We aim to find the drop locations that can be assigned to the uber driver based on maximum fare_amount per mile of distance covered. For this, we first use the Bellman ford Algorithm to find the shortest path from the source node i.e. “Grand Central Terminal” to all the nodes it is connected to in the weakly connected component subgraph. We further try to find the minimum average fare amount from the source to all the drop-off locations. We then compute a field – fare_amount/distance to get the

minimum fare per mile metric and sort it by descending order. The top drop-off locations give us the locations that can be prioritized by the company in assigning the pick-ups for that particular uber driver.

	distance	dropoff_location	fare_amount	fare_per_mile
2	0.449345	892 2nd Avenue Manhattan Community Board 6	17.00	37.832819
4	0.632547	Morgan Stanley Building 1585 Broadway	22.66	35.823412
1	0.361334	850 Third Avenue 850 3rd Avenue	10.00	27.675256
7	1.271566	67 Irving Place Manhattan Community Board 5	17.50	13.762559
5	0.677729	922 3rd Avenue Manhattan Community Board 6	6.50	9.590847
6	1.150330	59th Street - Columbus Circle (1) Broadway Man...	10.50	9.127816
12	1.847338	Papaya King 179 East 86th Street	15.50	8.390452
13	2.008945	181 West 10th Street Manhattan Community Board 2	14.50	7.217719
3	0.631652	Financial Times Building 1330 6th Avenue	4.50	7.124174
9	1.627706	East 79th Street & 3rd Avenue East 79th Street...	10.00	6.143614
11	1.732697	Heavenly Market & Deli East 11th Street Manhat...	9.00	5.194214
10	1.711833	647 Hudson Street Manhattan Community Board 2	8.50	4.965437
8	1.320024	Dante Alighieri West 63rd Street Manhattan Com...	6.50	4.924154
14	2.613963	153 East 99th Street Manhattan Community Board 11	10.00	3.825609
18	4.106360	1 West Street Financial District	15.00	3.652871
16	3.979652	107 East 126th Street East Harlem	12.00	3.015339
15	3.389166	Fulton Street Broadway Manhattan Community Boa...	8.50	2.507992
20	6.638544	1264 River Avenue High Bridge	16.00	2.410167

Based on the above results, we can recommend the prioritization of locations such as **2nd Avenue Manhattan Community Board 6, 1585 Broadway**, and **850 3rd Avenue** to be the priority destination for the uber driver pick-ups based on the maximization of fare per mile ratio when the pickup location is **"Grand Central Terminal"**. This method can be extended to other source locations as well.